

Imaginary-time density memory of a domain-wall parent ground state

Numerical benchmark notes

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We compute connected equilibrium density autocorrelations for the half-filled ground state of the signed overcomplete-Wannier parent. The calculation is deterministic and distinct from conditional purification or Born-averaged adaptive-circuit dynamics. Both local density memory and column-charge memory are obtained without many-body time stepping.

CONTENTS

I. Parent and state	1
II. Density fluctuations and spectral evaluation	1
III. Local and column observables	2
IV. Relation to filtering and numerical interpretation	2

I. PARENT AND STATE

With an overcomplete Wannier (OW) basis, define the single-particle parent

$$h_{\text{flat}} = \sum_R (P_{A+} + P_{B+} - P_{A-} - P_{B-}), \quad \hat{H} = \sum_{ij} (h_{\text{flat}})_{ij} \hat{c}_i^\dagger \hat{c}_j. \quad (1)$$

No subsequent replacement of energies by their signs is made. Distinct OW projectors need not commute. We retain hard support truncation, $N_x = 20$, $N_y = 30$, $\alpha_1 = 1$, $\alpha_2 = 30$, $n_{\text{shell}} = 1$, and trial orbital X . The spatial benchmark's regulator transforms representative OW columns at $k + 10^{-7}/N_y$ but retains the periodic untwisted Bloch basis. This is the occupation-regulator convention, not a flux-threading protocol. We diagonalize the resulting regulated parent in thirty 40×40 blocks and fill the lowest 600 modes. The same energies and eigenvectors define propagation. Write the ground state as $|\Omega\rangle$, its energy as E_0 , and $G_{ij} = \langle\langle \Omega | \hat{c}_i^\dagger \hat{c}_j | \Omega \rangle\rangle$. In the code, the occupied projector is $P = G^\text{T}$.

II. DENSITY FLUCTUATIONS AND SPECTRAL EVALUATION

For a Hermitian single-particle matrix D , let $\hat{O} = \sum_{ij} D_{ij} \hat{c}_i^\dagger \hat{c}_j$ and $\delta\hat{O} = \hat{O} - \langle\hat{O}\rangle\hat{1}$. For $\tau \geq 0$ the connected equilibrium correlator is

$$C_O(\tau) = \langle\delta\hat{O}(\tau)\delta\hat{O}(0)\rangle, \quad \hat{O}(\tau) = e^{\tau\hat{H}}\hat{O}e^{-\tau\hat{H}}, \quad (2)$$

$$C_O(\tau) = \langle\langle \Omega | \delta\hat{O} e^{-\tau(\hat{H}-E_0\hat{1})} \delta\hat{O} | \Omega \rangle\rangle. \quad (3)$$

A one-body insertion creates only particle-hole excitations. With a occupied and b empty this gives exactly

$$C_O(\tau) = \sum_{a \in \text{occ}, b \in \text{empty}} |u_b^\dagger D u_a|^2 e^{-\tau(\epsilon_b - \epsilon_a)}. \quad (4)$$

The weights and excitation energies are nonnegative. Therefore each autocorrelation is nonnegative and nonincreasing. At contact,

$$C_O(0) = \text{tr}[PD(\mathbf{1} - P)D]. \quad (5)$$

For total charge $D = \mathbf{1}$ this vanishes, as it must for a fixed-number Slater determinant. Imaginary time has inverse-parent-energy units; no identification with circuit cycles is assumed.

III. LOCAL AND COLUMN OBSERVABLES

Let $\hat{n}_{x,y} = \sum_{\mu} \hat{c}_{x,y,\mu}^{\dagger} \hat{c}_{x,y,\mu}$. We retain both

$$C_x^{\text{local}}(\tau) = \frac{1}{N_y} \sum_y \langle \delta \hat{n}_{x,y}(\tau) \delta \hat{n}_{x,y}(0) \rangle, \quad (6)$$

$$C_x^{\text{column}}(\tau) = \frac{1}{N_y} \langle \delta \hat{Q}_x(\tau) \delta \hat{Q}_x(0) \rangle, \quad \hat{Q}_x = \sum_y \hat{n}_{x,y}. \quad (7)$$

Translation symmetry makes the local origin average exact. Writing the Bloch eigenvectors as v_{ka} , local transition weights at fixed x are $|\sum_{\mu} v_{kb,x\mu}^* v_{k'a,x\mu}|^2 / N_y^2$. Column weights instead conserve momentum and are $\delta_{kk'} |\sum_{\mu} v_{kb,x\mu}^* v_{k'a,x\mu}|^2 / N_y$. The observables consequently probe different momentum sectors.

The existing spatial benchmark plots $B_x(r) = \frac{1}{2} \sum_{\mu\nu} |G_{(x,0,\mu),(x,r,\nu)}|^2$. Wick contraction gives, for distinct cells, $\langle \delta \hat{n}_{x,0} \delta \hat{n}_{x,r} \rangle = -2B_x(r)$. At $r = 0$ the contact term adds $\langle \hat{n}_{x,0} \rangle$. Thus the positive spatial proxy must not be mistaken for the signed density correlator, nor for the positive imaginary-time autocorrelation.

IV. RELATION TO FILTERING AND NUMERICAL INTERPRETATION

Normalized imaginary-time filtering evolves a state according to

$$\hat{\rho}(\tau) = \frac{e^{-\tau \hat{H}} \hat{\rho}(0) e^{-\tau \hat{H}}}{\text{Tr}[e^{-2\tau \hat{H}} \hat{\rho}(0)]}. \quad (8)$$

This operation alone does not define the two density insertions above. Furthermore, a finite ordered product of noncommuting postselected OW measurements is not automatically $e^{-\tau \hat{H}}$. The present benchmark contains neither such a measurement record nor an adaptive-channel average.

We save $\tau = 0$ and 240 logarithmically spaced positive times from 10^{-3} to 10^3 , using double precision throughout. Ratios $C_x(\tau)/C_x(0)$ are undefined when the equal-time variance is at most 10^{-12} ; raw data are never floored. There is no sampling SEM. A finite system has a discrete particle-hole spectrum, and a small gap matters only if its density matrix element is appreciable. No power-law exponent is fitted automatically.